

The Cyclic Sensitivity Floor

What came out of following the 3-6-9 doubling map into complexity theory. Everything below is either proven by exhaustive search or explicitly marked as unestablished. Two of my own results were withdrawn along the way; both retractions are recorded rather than quietly deleted.

The question

A Boolean function's **sensitivity** is the largest number of single-bit flips that can change its answer, taken over the worst input. Low sensitivity means stable.

Impose a symmetry and you constrain how low it can go:

- **Full symmetry** — the answer depends only on how many 1s the input has.
- **Cyclic symmetry** — the answer is unchanged by rotating the input. This is exactly the shift map the 3-6-9 doubling cycle turned out to be.

Cyclic is strictly weaker, so it should permit lower sensitivity. By how much is the question, and the large-n version of it is open.

Result 1 — the full-symmetry wall (exact, all n tested)

```
min sensitivity over non-constant symmetric functions = ceil((n+1)/2)
```

Verified exhaustively for $n = 3..16$, no exceptions. Full symmetry forces sensitivity to grow **linearly**. Symmetric functions can never be stable.

Result 2 — the cyclic floor (exact, $n = 3..12$)

[illegible]

full symmetry	2	3	3	4	4	5	5	6	6	7
gap	0	1	0	1	1	1	2	2	2	3

$n = 13$ did not finish within budget. No value is claimed for it.

The gap widens. Cyclic symmetry is not a constant improvement over full symmetry across this range — it pulls away as n grows.

Result 3 — the floor is not monotonic

$n = 9$ has floor **3**, lower than $n = 8$'s **4**. Adding a variable made stability *easier*.

The witness was independently verified: expanded to all 512 inputs, its sensitivity measured by definition, confirmed cyclically invariant and non-constant, and the $t = 2$ search confirmed exhausted. It is real.

The lowest-weight necklace it accepts is `001001001` — the period-3 string, and $9 = 3^2$.

The obvious explanation fails. If short shift-orbits caused the dip, $n = 8$ (divisors 2, 4) should dip harder than $n = 9$ (divisor 3 only). It does not. Divisor count does not explain this.

Whether dips recur at 25 and 27 is the natural test, and both are far beyond what exhaustive search can reach. So this stays an observation with one data point, which is not enough to be anything else.

Two things I got wrong

Withdrawn — annealing bounds. A first pass reported floors of 4, 5, 6, 8, 9 for $n = 8..12$. Re-running with ten times the compute lowered $n = 9$ from 6 to 5 and $n = 10$ from 7 to 6. Those numbers measured the search, not the mathematics. Discarded entirely. The exact solver later proved the true values are 4, 3, 4, 4, 4 — so the annealer was wrong by up to 5.

Fixed — a directional multiplicity bug. The first exact solver returned floor = 1 for every n , contradicting the verified census. Cause: in the necklace graph, adjacency multiplicity is **not symmetric**. From `000` three bit-flips reach `001`; from `001` only one returns. The solver credited both endpoints the same count. After the fix it reproduced all five known census values exactly.

Both errors were caught by checking against something independent. That is the only reason the table above can be trusted.

What this is and is not

Is: an exact, reproducible table; a verified non-monotonicity; a clean lemma (for cyclically invariant functions, non-degenerate = non-constant) that made the computation tractable; a measured widening gap.

Is not: new mathematics. $n \leq 12$ cannot distinguish the asymptotic behaviour that the open question is actually about. The large- n regime, where the real result would live, is untouched.

The honest summary: we followed a numerology claim to a real open problem, computed exactly as far as computation reaches, caught our own errors twice, and stopped where the evidence stopped.

Files

- `exact-floor.py` — the exact decision procedure
- `verify-floor.py` — independent brute-force verification of every value
- `the-floor2.py` — the two-sided symmetric-vs-cyclic comparison
- `the-floor.py` — the failed annealing attempt, kept for the record
- `backwards.py` — the antipodal observation and the bridge out of number theory